

Step 1 — Initial Process Specification

Process: Catalytic Pyrolysis of Plastic Waste

1. **Plastic Feedstock Type(s)** — The category of polymeric material used as input (e.g., PE, PP, PS, PET, or mixed plastic waste). Feedstock identity determines degradation mechanisms and strongly influences product distribution.
2. **Feedstock Composition** — The relative proportion of polymer types present in the input stream. Composition affects selectivity, reproducibility, and comparability across experimental studies.
3. **Catalyst Type** — The catalytic material applied during the process (e.g., zeolites, FCC catalysts, metal-supported systems). Catalyst structure and chemistry govern cracking pathways and product selectivity.
4. **Catalyst-to-Feed Ratio** — The mass ratio between catalyst and plastic feedstock. This parameter controls conversion efficiency and distribution of liquid, gas, and solid fractions.
5. **Reactor Configuration** — The design and operational setup of the reactor (e.g., fixed-bed, fluidized-bed, batch, continuous). Reactor configuration influences heat transfer, mass transfer, and scalability.
6. **Reaction Temperature** — The operating temperature at which thermal degradation occurs, typically reported in °C. Temperature is a primary driver of reaction kinetics and product yield distribution.
7. **Heating Rate** — The rate at which temperature is increased during the process. Heating rate affects polymer breakdown pathways and the formation of secondary products.
8. **Residence Time** — The duration for which the feedstock and intermediates remain under reaction conditions. Residence time influences conversion extent and secondary cracking reactions.
9. **Reaction Atmosphere** — The gaseous environment within the reactor (e.g., nitrogen, argon, hydrogen). The atmosphere determines oxidative conditions and can modify reaction pathways.
10. **Operational Scale** — The experimental scale at which the process is conducted (laboratory, pilot, or industrial). Scale impacts reproducibility, comparability, and technological relevance.

11. **Liquid Product Yield** — The proportion of liquid fraction generated during pyrolysis, typically expressed as a percentage of initial mass. Often used as a primary performance indicator.
12. **Gaseous Product Yield** — The proportion of gaseous products formed during the reaction, reported as a percentage of initial mass.
13. **Solid Residue Yield (Char)** — The proportion of solid carbonaceous residue remaining after conversion. Char formation reflects incomplete degradation and catalyst effects.
14. **Product Composition** — The main chemical compounds identified in liquid and gas fractions. Product composition is typically determined via chromatographic and spectroscopic analysis.
15. **Analytical Characterisation Methods** — The experimental techniques used to analyse feedstock and products (e.g., GC-MS, FTIR, TGA). These methods ensure reproducibility and enable cross-study comparison.